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The Neoproterozoic climatic paradox: Equatorial palaeolatitude for Marinoan glaciation near sea level in South Australia

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Abstract

New palaeomagnetic analyses have been carried out for the Neoproterozoic (650–600 Ma) Elatina Formation, an important redbed unit of the Marinoan glaciogenic sequence in the Adelaide Geosyncline, South Australia, and flat-lying equivalent facies on the adjacent cratonic Stuart Shelf and Torrens Hinge Zone. The Marinoan rocks display strong evidence of marine glacial deposition, and coeval periglacial sand wedges in permafrost regolith on the Stuart Shelf indicate in-situ cold climate near sea level and marked seasonality. The palaeomagnetic data define a palaeopole for the formation and indicate that Marinoan glaciation, including permafrost, grounded glaciers and marine glacial deposition, occurred near the palaeoequator.

Rocks analysed include 97 oriented outcrop samples from 15 sites at three widely separated sections (~ 65–115 m thick) of gently folded and unmetamorphosed sandstone, siltstone and tillite spanning the Elatina Formation in the Central Flinders Zone of the Adelaide Geosyncline, soft-sediment folds from tidal rhythmites, and 60 specimens from 54 core samples from six deep drillholes on the Stuart Shelf and Torrens Hinge Zone. The most stable remanence components were only completely demagnetised by 685°C, indicating that haematite is the likely carrier of the remanent magnetisation. This conclusion is supported by the presence of ultrafine haematitic pigment coating clastic grains and filling interstices in the rocks. The observation of mixed polarities within some sandstone samples suggests that such lithologies acquired their remamence as chemical remanent magnetisation (CRM). A positive fold test on syndepositional soft-sediment folds in tidal rhythmites confirms that the rhythmites acquired a detrital remanent magnetisation (DRM) by the settling of haematite grains from suspension in quiet waters. Concordant palaeomagnetic directions determined for the rhythmites and other facies of the Elatina Formation show that the formation acquired its CRM close to the time of deposition. The existence of polarity reversals within a stratigraphic section and within some samples therefore argues strongly for the identification of a dipole field axis and the sufficient averaging of secular variation to define a palaeopole for the formation. The palaeopole derived from the oriented sample results (structurally corrected) is located at 52.4°S, 347.1°E ($d_p = 3.7°$, $d_m = 7.4°$); this pole position is consistent with the Neoproterozoic apparent polar wander path for Australia. The overall formation mean direction determined from the sample results (structurally corrected) has a declination of 197.3° and an inclination of −5.3° ($\alpha_{95} = 7.4°$), and indicates a mean palaeolatitude of deposition of 2.7° ± 3.7°N. These results accord with the virtual geomagnetic pole previously determined for tidal rhythmites of the Elatina Formation and provide the strongest evidence yet for the equatorial palaeolatitude of Neoproterozoic glaciation.

The palaeomagnetic results, together with geological observations for Marinoan glaciogenic rocks, therefore confirm the Neoproterozoic (pre-Ediacaran, ≥ 590 Ma) climatic paradox in South Australia: *frigid strongly seasonal climate, permafrost, and grounded glaciers near sea level in equatorial palaeolatitudes.* Resolution of this paradox may illuminate Precambrian planetary dynamics and the change in global state during the Ediacaran.

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1. Introduction

1.1. Regional stratigraphy and age of the Marinoan glaciogenic sequence

The Adelaide Geosyncline region in South Australia (Fig. 1), which contains one of the most complete and well preserved Neoproterozoic successions in the world, provides clear evidence of two major glaciations during the Neoproterozoic [1,2]: The Sturtian glaciation between about 800 and 750 Ma and the Marinoan glaciation between 650 and 600 Ma (Fig. 2). ('Adelaide Geosyncline' is retained in South Australia "as a formal term in the broad and non-genetic sense of geosyncline as originally defined, i.e. a complex basin of very thick sediment accumulation" [1].) Here we are concerned with the Marinoan glaciogenic sequence, of which the arenaceous and tillitic Elatina Formation is an important stratigraphic unit [1–3]. Indeed Mawson [4], following his recognition of tillite in the Elatina Formation, proposed the name 'Elatina glaciation' for this younger Neoproterozoic glacial event.

The Elatina Formation was deposited widely in the western, central and northeastern parts of the Adelaide Geosyncline (Fig. 1), and consists mainly of reddish brown siltstone, fine-to coarse-grained sandstone, conglomerate, and tillite containing faceted, polished and striated clasts. The formation commonly attains a thickness of between 100 and 500 m. The arenaceous facies accumulated in fluvioglacial to marine-shelf environments, and locally includes tidal rhythmites that have provided a key datum for the Earth's palaeorotation [5–8]. To the north and southeast the Elatina Formation passes into thick (1000–> 1500 m) basinal deposits of sandstone, tillite and siltstone with dropstones that constitute the Yerelina Subgroup. On the Stuart Shelf, the cratonic platform west of the Adelaide Geosyncline, equivalents of the Elatina Formation comprise pale red to pinkish white sandstone (periglacial–aeolian Whyalla sandstone) and reddish brown siltstone, mudstone and diamictite in the subsurface. In most areas the Elatina Formation is overlain by a distinctive marker laminated dolostone, the Nuccaleena Formation. As concluded by Lemon and Gostin [3], the presence in the Elatina Formation of a basal tillite and a dropstone diamictite in contact with the overlying dolostone demonstrates that the entire formation was deposited under glacial conditions. Indeed, much field evidence indicates that a frigid climate prevailed near sea level during Marinoan glaciation in South Australia (see Section 3.1).

No geochronological data are available for the Marinoan glaciogenic sequence and the age of glaciation must be estimated indirectly. The Elatina Formation lies stratigraphically well below the level of first appearance of Ediacaran megafauna at the top of the Wonoka Formation (Fig. 2) in the Adelaide Geosyncline [9]. The Yarloo Shale on the Stuart Shelf, the equivalent of the Bunyeroo Formation in the Geosyncline (Fig. 2), gave a Rb–Sr whole-rock isochron age of 588 ± 35 Ma [1,2]; combined Rb–Sr whole-rock data for the Yarloo Shale and subjacent shales gave a mean age of 593 ± 32 Ma [10]. Thus, the age of the Yarloo Shale and Bunyeroo Formation may be approximated as 590 Ma, which sets a younger limit for the age of Marinoan glaciation. Other Rb–Sr whole-rock shale isochron ages [1,2] relevant to the age of Marinoan glaciation include a partially reset age of 601 ± 68 Ma for the Brachina Formation on the Curnamona Craton east of the Adelaide Geosyncline, and 724 ± 40 Ma for strata of the 'Yudnapinna Beds' and 750 ± 53 Ma for the Tapley Hill Formation on the Stuart Shelf; the latter formation directly overlies the Sturtian glaciogenic sequence and is the principal 'interglacial' formation in the Adelaide Geosyncline region. While recognising that such Rb–Sr whole-rock ages for shales must be interpreted with caution, available geochronological and stratigraphic information suggests that Marinoan glaciation took place between 650 and 600 Ma.

1.2. Previous palaeomagnetic studies of the Elatina Formation

Previous palaeomagnetic studies of Neoproterozoic strata in the Adelaide Geosyncline, including the Sturtian Merinjina Tillite, the Tapley Hill Formation and the Elatina Formation, have suggested that Neoproterozoic glacial deposition and periglacial activity occurred in low ($\leq 12°$) palaeolatitudes [11–14]. Palaeomagnetic data for the tidal rhythmite member of the Elatina Formation at Pichi Richi Pass [12] indicated a mean structurally corrected inclina-

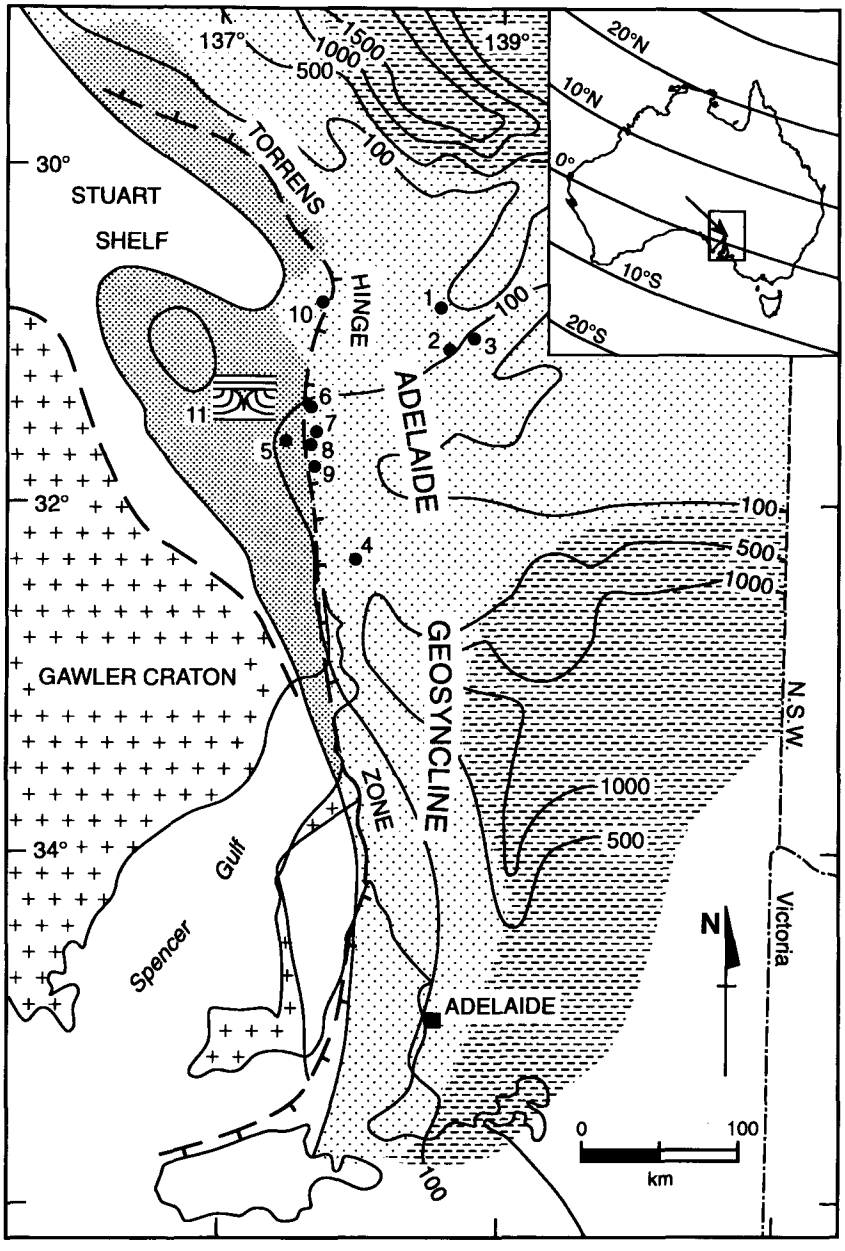


Fig. 1. Sedimentary facies of the Marinoan glaciogenic sequence in the Adelaide Geosyncline region, South Australia (adapted from [1,2]), and localities of field sections and drillholes sampled for the present study. Dense stipple = mainly periglacial–aeolian Whyalla Sandstone, and Elatina equivalent in the subsurface. Light stipple = Elatina Formation, shallow-water marine sandstone and siltstone, fluvioglacial sandstone, and tillite and dropstone facies. Dashes = Yerelina Subgroup, basinal sandstone, tillite, and siltstone with dropstones. Isopachs in metres. Field sections in the Adelaide Geosyncline: *1* = Mooloolloo; *2* = Bulls Gap; *3* = Second Plain; *4* = Pichi Richi Pass. *5–10* = Drillholes on the Stuart Shelf and Torrens Hinge Zone (see text). *11* = Periglacial structures in Marinoan permafrost regolith and in the Whyalla Sandstone on the Stuart Shelf (Fig. 7). The inset shows palaeolatitudes for Australia during Marinoan glaciation, based on the present study; the arrow represents the mean palaeowind direction determined for the periglacial–aeolian Whyalla Sandstone on the Stuart Shelf [31].

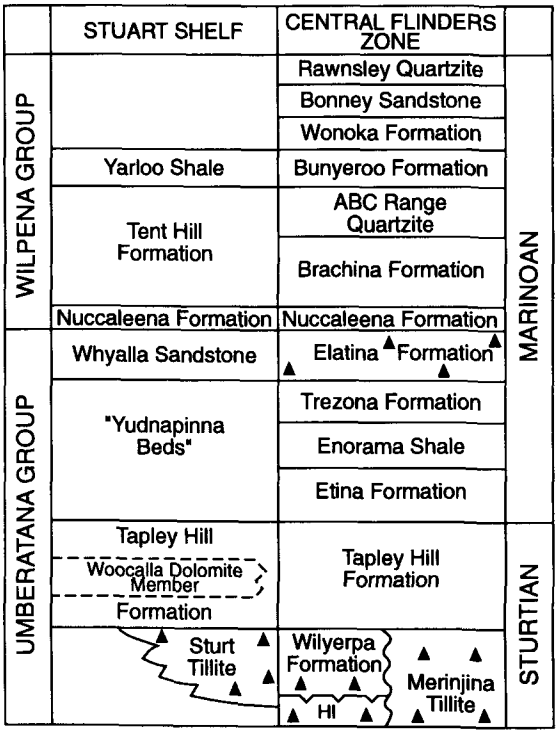


Fig. 2. Simplified stratigraphy of the late Neoproterozoic Umberatana and Wilpena groups of the Adelaide Geosyncline (Central Flinders Zone) and Stuart Shelf. Formations containing tillite are shown with triangles. *HI* = Holowilena Ironstone. The Wilpena Group is disconformably overlain by Early Cambrian strata, and the Umberatana Group overlies the Neoproterozoic Burra Group with regional unconformity (see [1,2]).

tion of $-9.6°$. The magnetisation of the tidal rhythmites is extremely stable, typically showing appreciable decay only after heating above 660°C. Secondary overprinting is not evident. The magnetic carriers are mainly silt-grade detrital grains of haematite and martite of high temperature, igneous or metamorphic origin; magnetic overprinting of such detrital carriers would require metamorphism under conditions appropriate to at least upper amphibolite facies. Because the rocks sampled have undergone only burial diagenesis at low temperatures (< 135–160°C), Embleton and Williams [12] concluded that the remanence must be primary and date closely to the time of deposition.

These findings were supported by positive fold tests on small folds, demonstrably of soft-sediment origin, displayed by the tidal rhythmites at Pichi Richi Pass [13–15]. The detailed study by Schmidt et al. [13,14] confirmed that the remanence is indeed primary and was acquired very soon after deposition, prior to and/or during formation of the folds. Very early acquisition of a detrital remanent magnetisation (DRM) for this facies was explained by the magnetic mineralogy and envisaged mechanism of deposition —settling of detrital haematite grains from suspension in quiet waters below storm wave base on a distal ebb-tidal delta [5–8]. The Elatina results agreed with the mean palaeomagnetic pole for South Australia for the late Neoproterozoic, which with other palaeomagnetic data places the Adelaide Geosyncline at a palaeolatitude of 10–12°N.

Meert and Van der Voo [16] have recently questioned the reliability of palaeomagnetic data for the Elatina Formation, suggesting the small-scale folds in the tidal rhythmites that were studied palaeomagnetically may be tectonic and related to the Delamerian Orogeny that deformed the Adelaide Geosyncline in Cambro-Ordovician time. If the folds were shown to be Delamerian, the fold tests would not provide a tight constraint on the timing of remanence in the Elatina Formation. In response, Williams et al. [17] presented unequivocal sedimentological evidence that the folds are indeed of soft-sediment origin: erosional and depositional crestal features clearly show that the folds studied by Schmidt et al. [13,14] were in the early stages of development at the sediment surface during deposition. The soft-sediment structures in the Elatina rhythmites are interpreted as seismites and related gravity slumps and slides induced by seismic activity during deposition. The acquisition of a DRM was followed by further soft-sediment folding.

However, as emphasised by Schmidt et al. [13,14] and reiterated by Meert and Van der Voo [16], the Elatina pole is a virtual geomagnetic pole (VGP) because the rhythmites studied are of tidal origin and represent only 60–70 years of deposition [5–8]. Schmidt et al. [13] and Williams et al. [17] noted that the Elatina VGP position is similar to that of other Neoproterozoic palaeopoles for the Adelaide Geosyncline [11] and for the Amadeus Basin in central Australia [18] and accords with cleaned directions and inclinations for Neoproterozoic strata in the Ngalia and Georgina basins farther north [19]. They concluded that the low inclination obtained for the

Elatina rhythmites does not record a geomagnetic excursion or transition, or significant inclination error. Nonetheless, palaeomagnetic studies of the Elatina Formation must be viewed as incomplete in that secular variation had not been adequately averaged.

The present study aims at rectifying this deficiency by providing further palaeomagnetic data spanning the Elatina Formation in the Adelaide Geosyncline and flat-lying equivalent facies on the Stuart Shelf and Torrens Hinge Zone to define a palaeopole and a palaeolatitude for the formation as a whole. These new data affirm our previous conclusion of a palaeoequatorial setting for Marinoan glaciation in South Australia.

2. Palaeomagnetic analysis of the Elatina Formation

The present study combines several separate palaeomagnetic analyses of samples from the Elatina Formation. Analyses have been carried out on 97 oriented outcrop samples from three stratigraphic sections spanning the formation in the Adelaide Geosyncline, an oriented block sample displaying soft-sediment slump folds from Pichi Richi Pass, and 60 specimens from 54 core samples from six deep drillholes in flat-lying Elatina equivalent on the Stuart Shelf and Torrens Hinge Zone.

2.1. Techniques

Routine palaeomagnetic methods [20,21] were used throughout. Field samples were collected using a hand-held drill and oriented with a magnetic compass and whenever possible also with a sun compass. Both alternating field (AF) and thermal demagnetisation were used, although the latter proved to be more effective and was therefore generally employed. The demagnetisers used were a Schonstedt AF demagnetiser model GSD-1 and the CSIRO three-stage carousel furnace. The furnace is housed inside a 4 m ten-coil Helmholtz set with automatic feedback maintaining zero-field (< 5 nT). Remanent magnetisations were measured using a semi-automated CTF three-axis cryogenic magnetometer. Changes in magnetic susceptibility during thermal demagnetisation were monitored by using a weak DC field (~ 300 nT) trapped in the magnetometer.

Directions of remanent magnetisation components were determined using a WINDOWS® version of LINEFIND [22] for principal component analysis (PCA). The linear segments are fitted to data points weighted according to the inverse of their measured variances. The variances are represented in plots of orthogonal projections by sized dots.

2.2. Outcrop in the Adelaide Geosyncline

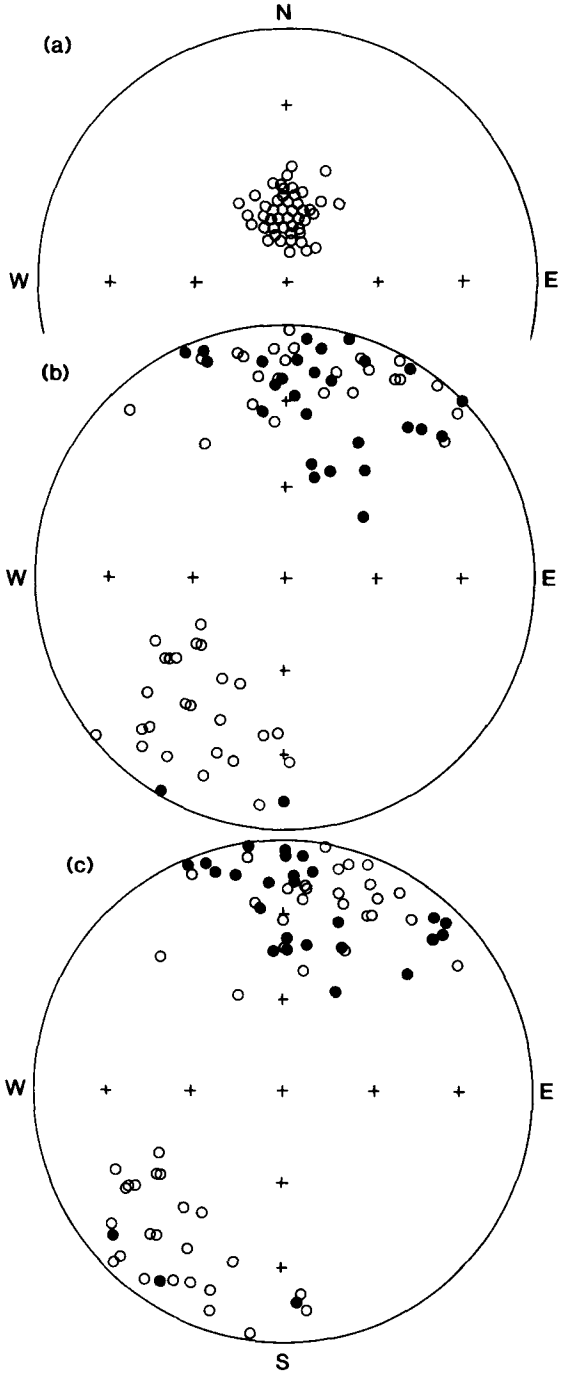
Stratigraphy and mineralogy

Outcrops of the Elatina Formation sampled for palaeomagnetic analysis were restricted to the Central Flinders Zone [1] between latitudes 31°00′ and 31°30′S and west of longitude 139°00′E, to avoid the effects of local tight folding and slaty cleavage related to the Delamerian Orogeny. The Central Flinders Zone is the least-deformed province of the Adelaide Geosyncline, being characterised by broad, dome-and-basin folds. The sediments display no obvious sign of metamorphism and the low degree of deformation approaches that of the near-horizontal Neoproterozoic sequence on the Stuart Shelf and western Torrens Hinge Zone.

A total of 97 fully oriented core samples were taken from outcrops at 15 sites, spanning three widely separated sections of the Elatina Formation. The sections sampled are well exposed in creek beds at Mooloolo (31°01′S, 138°34′E; Sites 01–05, numbers increase down-section), Bulls Gap (31°15′S, 138°38′E; Sites 06–08), and Second Plain (31°10′S, 138°47′E; Sites 09–15). These are sections 15, 10 and 30, respectively, of Lemon and Gostin [3]. Strata at the sections sampled are gently dipping and show no cleavage or other evidence of metamorphism. Section thicknesses and mean attitudes of bedding are ~ 65 m and strike 36°, dip 19°NW (Mooloolo), ~ 90 m and strike 145°, dip 15°SW (Bulls Gap), and ~ 115 m and strike 153°, dip 15°NE (Second Plain). Because of lack of suitable outcrop, samples were not obtained from areas with strikes close to E–W.

The main lithologies of the Elatina Formation [3] were sampled. These are pale red to pale reddish brown (10R 5-6/2-4), cross-bedded and slumped medium-to coarse-grained sandstone with granule bands, overlain by reddish brown (10R 4-5/3) silt-

stone with dropstones and interbeds of reddish brown tillite, then reddish-brown ripple cross-laminated and poorly bedded very fine- to fine-grained sandstone with local heavy mineral bands.



Magnetic minerals in the samples comprise detrital opaque grains and reddish brown haematitic pigment. The opaque grains range from silt to fine sand grade and are usually scattered throughout the rocks, but in some of the ripple cross-laminated sandstones are concentrated in thin bands. The detrital opaques in the sandstones consist mainly of haematite showing variable alteration to goethite; detailed study of separated detrital opaques from the rhythmites at Pichi Richi Pass [12] identified grains of titaniferous haematite, martite (haematite) after magnetite, and haematite and 'leucoxene' after ilmenite. The haematitic pigment is ultrafine and occurs as a coating on detrital quartz and other grains as well as filling interstices between grains. Total iron content (expressed as percent Fe_2O_3) is influenced by lithology: two samples of tillite matrix assayed 3.59 and 3.69% Fe_2O_3, nine samples of rippled very fine-to fine-grained sandstone averaged 2.98% (range 2.43–3.99%), and five samples of medium-grained sandstone averaged 1.71% (1.27–2.01%) (all assays by X-ray fluorescence). Much of the iron content of the rocks is contained in the haematitic pigment.

Results

Natural remanent magnetisation (NRM) intensities averaged 4.85 mA m^{-1}, with a standard deviation of 3.13 mA m^{-1}. The mean NRM direction has a declination of 355° and an inclination of −64°. Fig. 3a shows directions of the magnetisation component demagnetised below 400°C, the mean direction of which has a declination of 0.9° and an inclination of −67.8° ($\alpha_{95} = 1.4°$). In comparison, the present field direction for the sample area has a declination of 7° and an inclination of −64°, and a dipole field declination of 0° and an inclination of −50°. Clearly the remanence of the Elatina Formation from the three sections sampled is dominated by a late Tertiary present field component. This is dis-

Fig. 3. Equal-area stereographic projections of combined palaeomagnetic data for oriented samples of the Elatina Formation from three stratigraphic sections (Bulls Gap, Mooloolo, Second Plain) in the Adelaide Geosyncline, showing (a) low-temperature (< 400°C) component directions, and high-temperature (> 650°C) component directions before (b) and after (c) structural correction. • = Lower hemisphere; ◦ = upper hemisphere.

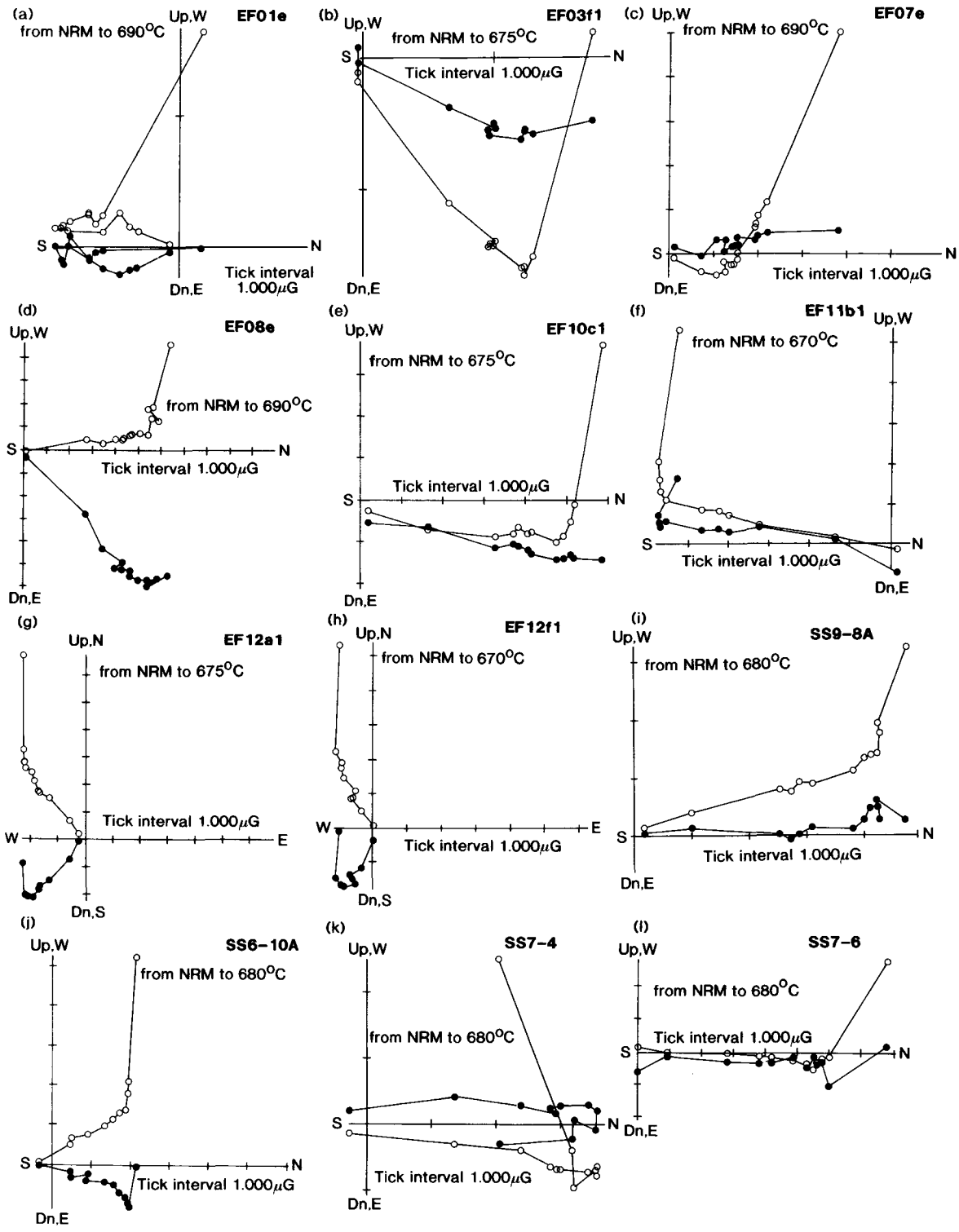


Fig. 4. Orthogonal projections showing examples of behaviour on thermal demagnetisation for eight fully oriented samples of the Elatina Formation from three stratigraphic sections (Bulls Gap, Mooloolo and Second Plain) in the Adelaide Geosyncline (a–h), and four specimens from core samples from deep drillholes on the Stuart Shelf–Torrens Hinge Zone (i–l). The sample identification includes the site or drillhole number. Steps shown are NRM, 200°C, 300°C, 400°C, 500°C, 530°C, 550°C, 600°C, 625°C, 650°C, 665°C, 675°C, 680°C and 685°C (some steps omitted after complete demagnetisation). • = Magnetisation vectors projected onto the horizontal plane; ◦ = projections onto the vertical plane; *Dn* = down. Units are mA m^{-1} (= 1 mG = 10^{-6} emu/cc).

tinctly different from the NRM of the Elatina rhythmites sampled at Pichi Richi Pass, which generally possessed a very minor amount of present-field-related remanence [12].

The predominance of recent remanence components is apparent in plots of the demagnetisation results, which show well-developed northerly and upwardly directed remanence components demagnetised by 400°C, leaving more stable remanent components which only demagnetise at higher temperatures (Fig. 4a–h). In some samples two higher temperature components are discernible. These components are almost anti-parallel, probably reflecting a polarity reversal of the geomagnetic field during the magnetisation process. For example, stepwise demagnetisation of sample EF01e reveals first a present field component, secondly a northerly directed component, and last a southerly directed component (Fig. 4a). In most samples the remanence at higher temperatures comprises a single well-defined component directed generally to the north or to the south (see plots of results from EF10c1 and EF11b1, Fig. 4e and 4f).

The southerly directed components are very close to the mean cleaned remanence direction determined by Embleton and Williams [12] for the rhythmite member at Pichi Richi Pass. For the purposes of calculating mean directions, only one polarity was assigned to each sample because usually one component predominated and could be defined with greater precision by PCA. The magnetic polarity at most sites was predominantly of a single polarity although four sites clearly possessed dual polarity, and as discussed above, dual polarity was exhibited by some samples. Of the ten sites which yielded internally consistent results, five are overwhelmingly of normal polarity (northward), two are of reversed polarity and the remaining three are of mixed polarity. It is possible to define a rough magnetostratigraphy only at the Second Plain section, where polarity changes from normal (Site 10) to reversed (Sites 11 and 12) down-section. Only normal and mixed polarities were encountered at the other two sections.

The most stable remanence components displayed by the Elatina Formation were only completely demagnetised by 685°C, indicating that haematite is the likely carrier of the remanent magnetisation. This finding is consistent with the petrographic evidence discussed above. The observation of mixed polarities within some sandstone samples suggests that such lithologies of the Elatina Formation acquired their remanence as chemical remanent magnetisation (CRM), with each sample representing different time intervals and different averaging of the secular variation. In these circumstances it is appropriate to combine sample directions, rather than site directions, to provide an overall formation mean direction. The Elatina Formation in the three sections studied appears to be most consistent with the Type A redbed classification of Turner [23], for which the mineralogy at the time of deposition is fairly mature and remanence is an early CRM.

Fig. 3b and 3c show stereographic projections of the high-temperature (> 650°C) components, both in situ and after structural correction. Table 1 lists the high-temperature in-situ mean site directions, the in-situ and structurally corrected site mean directions, and the overall structurally corrected formation mean direction derived from the sample results, which has a declination of 197.3° and an inclination of -5.3° ($\alpha_{95} = 7.4$°). The latter direction compares very closely to the structurally corrected result of Embleton and Williams [12], namely a declination of 191.9° and an inclination of -9.6° ($\alpha_{95} = 3.4$°).

Following structural correction of high-temperature components, the sample (Fig. 3b and c) and site mean directions are more clustered visually and the precision parameter marginally improves from 10.8 to 11.1 (Table 1). However, the strike directions of the strata are not sufficiently different from the remanence directions and/or the bedding dips are too low to yield a definitive fold test. The McFadden [24] fold test examines the correlation between magnetisation directions and the dip directions. A minimum correlation (lowest SCOS value) yields the most likely attitude of the strata at the time of magnetisation. A high correlation (largest SCOS value) suggests that the magnetisation has been rotated with the bedding and therefore pre-dates the folding. Because the mean site direction is little changed before and after correcting for bedding attitude, the appropriate formulation of McFadden's [24] fold test follows his 'definition 1'. Using this, no apparent correlation exists between the displacement of site mean directions from the mean site directions and their structural corrections. Both the in-situ SCOS

statistic and its value after structural correction are less than the 95% critical value for ten sites. Hence the fold test is inconclusive, although the power of such a tectonic fold test in constraining the timing of remanence is subordinate to that of the soft-sediment fold test previously reported [13,14] and enhanced below, because regional folding in the Adelaide Geosyncline occurred more than 100 m.y. after deposition of the Elatina Formation.

Palaeomagnetic data for the underlying Tapley Hill Formation and overlying Bunyeroo Formation [11], together with more recent data, indicate that

Table 1
Elatina Formation high-temperature (> 650°C) in-situ mean site directions for oriented samples, in-situ and structurally corrected site mean directions, and structurally corrected sample mean direction, for three stratigraphic sections (Bulls Gap, Mooloolo and Second Plain) in the Adelaide Geosyncline

	N	Dm (°)	Im (°)	k	α_{95} (°)
Site 01	11	4.2	-9.9	13.4	12.9
Site 03	6	13.2	40.7	52.6	9.3
Site 04	5	194.8	1.6	6.60	32.2
Site 05	8	1.9	-20.0	4.89	27.9
Site 07	6	4.4	-0.7	17.2	16.7
Site 08	5	28.7	24.8	9.56	26.1
Site 10	12	13.0	-8.1	12.5	12.8
Site 11	9	227.6	-21.5	9.78	17.3
Site 12	11	216.8	-26.4	26.8	9.0
Site 13	6	13.9	-3.0	6.31	28.9
Site mean (in situ)	10	196.9	-13.3	10.84	15.4
Site mean (corrected; see Section 2.2.1 for structure at sites)	10	197.4	-7.1	11.07	15.2
Palaeomagnetic pole: latitude = 51.5°S, longitude = 346.6°E (dp = 7.7, dm = 15.3)					
Sample mean (corrected; see Section 2.2.1 for structure at sites)	79	197.3	-5.3	5.68	7.4
Palaeomagnetic pole: latitude = 52.4°S, longitude = 347.1°E (dp = 3.7, dm = 7.4)					
cf. Embleton and Williams [12]		191.9	-9.6		3.4

N = Number of samples and sites; D_m = mean declination; I_m = mean inclination; κ = precision parameter; α_{95} = semi-angle of cone of 95% confidence about mean direction; d_p, d_m = semi-major and semi-minor axes of polar error ellipse (°)

these formations have preserved low-inclination remanence components that pre-date tectonic folding. The fold test applied by McWilliams and McElhinny [11] to the Tapley Hill Formation was marginally below the 95% confidence level but was positive at the 90% level for the stable component with a structurally corrected mean declination of 47° and mean inclination of $-9°$ ($\alpha_{95} = 15°$). The initial fold test for the Bunyeroo Formation [11] suffered from too little variation in structural attitude. Recent work on the Bunyeroo Fortmation has addressed this deficiency through sampling fold limbs with steep dips (60–63°) and strikes at large angles to the remanence directions. The recent findings provide very strong evidence for pre-folding magnetisation using McFadden's [24] fold test (significant at the 99% confidence level); the stable, high-temperature component has a structurally corrected mean declination of 54.9° and mean inclination of 28.2° ($\alpha_{95} = 3.6°$) [P.W. Schmidt, unpublished]. The identification of pre-folding remanent components with low inclinations in late Neoproterozoic sedimentary units stratigraphically above and below the Elatina Formation (Fig. 2) argues strongly that the stable low-inclination component in the Elatina Formation likewise pre-dates tectonic folding.

The Elatina data also have been subjected to a reversal test (see [25]). The mean of the 52 normal

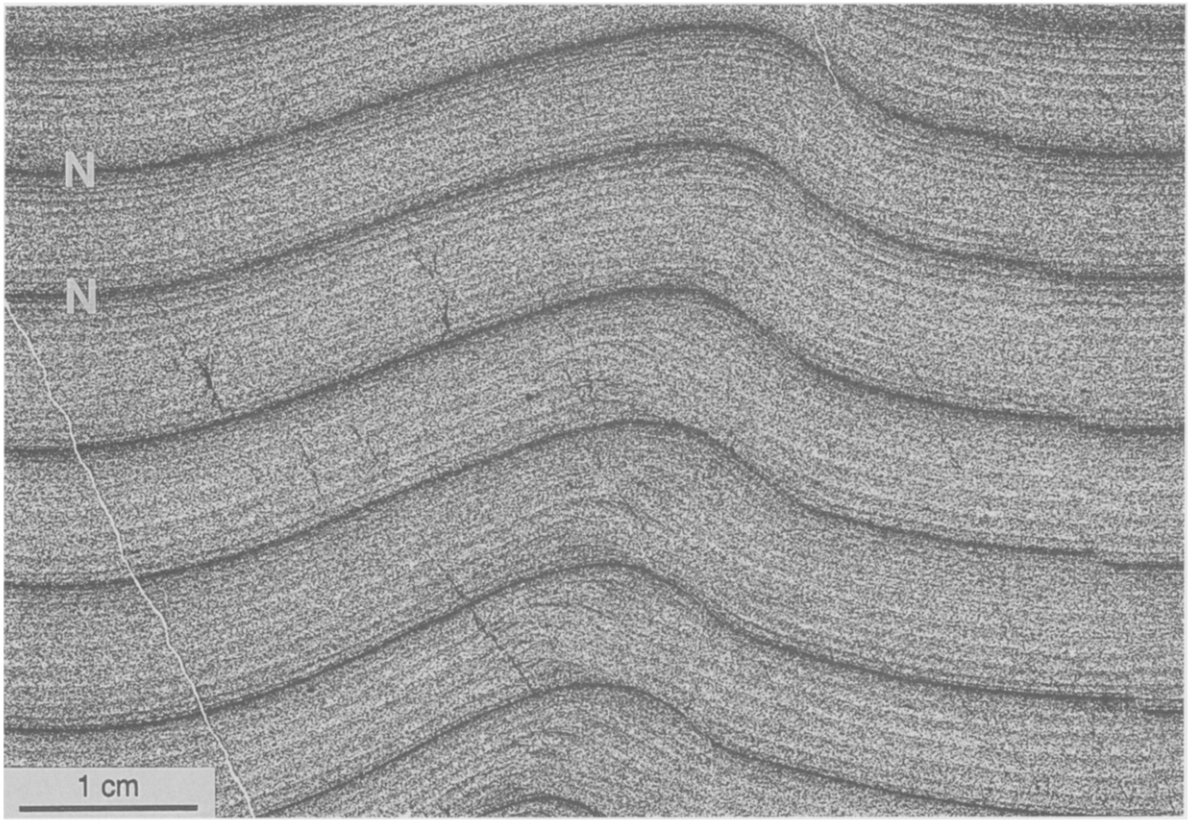


Fig. 5. Tidal rhythmites of the Elatina Formation at Pichi Richi Pass, showing an E–W section of part of the soft-sediment fold structure studied for the mini-specimen fold test. The photograph spans seven vertically accreted fortnightly (neap–spring) cycles of diurnal (lunar day) graded laminae of fine sand to silt grade. The dark bands (N) delineating the cycles are drapes of fine-grained haematitic clayey material deposited during slack water at neap tides. During deposition of the laminae by ebb-tidal currents, delicate wedge-outs and truncations occurred at the fold crest. These features indicate that the fold was partly developed at the sediment surface during deposition and is of soft-sediment origin. The block sample used for the fold test contains twenty fortnightly cycles and so was deposited in less than one year. Thin section viewed with transmitted light. Scale bar = 1 cm.

directions has a declination of 9.7°, an inclination of −1.2° and a resultant of 44.452, whereas the mean of the 27 reversed directions has a declination of 214.8°, an inclination of −20.0° and a resultant of 24.487. The calculated critical angle of 12.3° is less than the angle of 32.5° (mainly in inclination) between the normal and reversed axes, indicating a negative reversal test. The significance of this is not pursued further here. It is noteworthy, however, that similar asymmetries have been identified in the Proterozoic geomagnetic record that cannot simply be dismissed as artefacts or the effects of remagnetisation but that may reflect the morphology of the Proterozoic geomagnetic field [26]. Discernible asymmetry in normal and reverse inclinations is also evident in palaeomagnetic data for the past 5 m.y. and has been attributed to lateral variations in temperature or composition at or near the core–mantle boundary [27].

2.3. Soft-sediment mini-specimen fold test, Pichi Richi Pass

An oriented block sample of typically well-laminated tidal rhythmite displaying conjugate soft-sediment folds trending E–W (principal fold) and N–S (Fig. 5) was collected from an outcrop of the 10 m thick rhythmite member at Pichi Richi Pass (32°25′S, 137°59′E) near the western margin of the Adelaide Geosyncline. This sample came from the same site as the soft-sediment folds previously studied by Schmidt et al. [13,14]. The petrography and geochemistry of the rhythmites were discussed by Embleton and Williams [12]. As noted above, Williams et al. [17] presented sedimentological evidence such as crestal truncations and lee-side sand lenses indicating that such small folds in the Elatina Formation are of soft-sediment origin and were in the early stages of development at the sediment surface during deposition. Delicate wedge-outs and truncations of laminae at fold crests in the present sample (Fig. 5) confirm this finding.

In our previous investigation of a sample with soft-sediment folds from Pichi Richi Pass [13,14], standard-sized palaeomagnetic core specimens (25 mm diameter × 22 mm height) were used; these specimens were larger than the fine-scale fold structure shown by the sample, and some individual specimens encompassed a wide range of bedding attitudes (as much as 5–10° of dip). For the present investigation we used core mini-specimens (9 mm diameter × 8 mm height) to gain better resolution of

Table 2
Fold test [24] for 185 mini-specimens from the oriented block sample displaying soft-sediment folds from Pichi Richi Pass. Data are for high-temperature (> 650°C) directions structurally corrected for tectonic dip

Group	% unfolding	Dm (°)	Im (°)	k	α_{95} (°)	SCOS[a]
Before unfolding	0	190.0	−9.1	22.9	2.2	78.669
Unfolded	100	190.0	−7.5	28.9	2.0	42.326
Max k	66	190.1	−8.1	32.3	1.9	5.429
Min SCOS	68	190.1	−8.0	32.2	1.9	0.699

[a] The lowest SCOS value yields the most likely attitude of the strata at the time of magnetisation (see text for discussion). Symbols as for Table 1

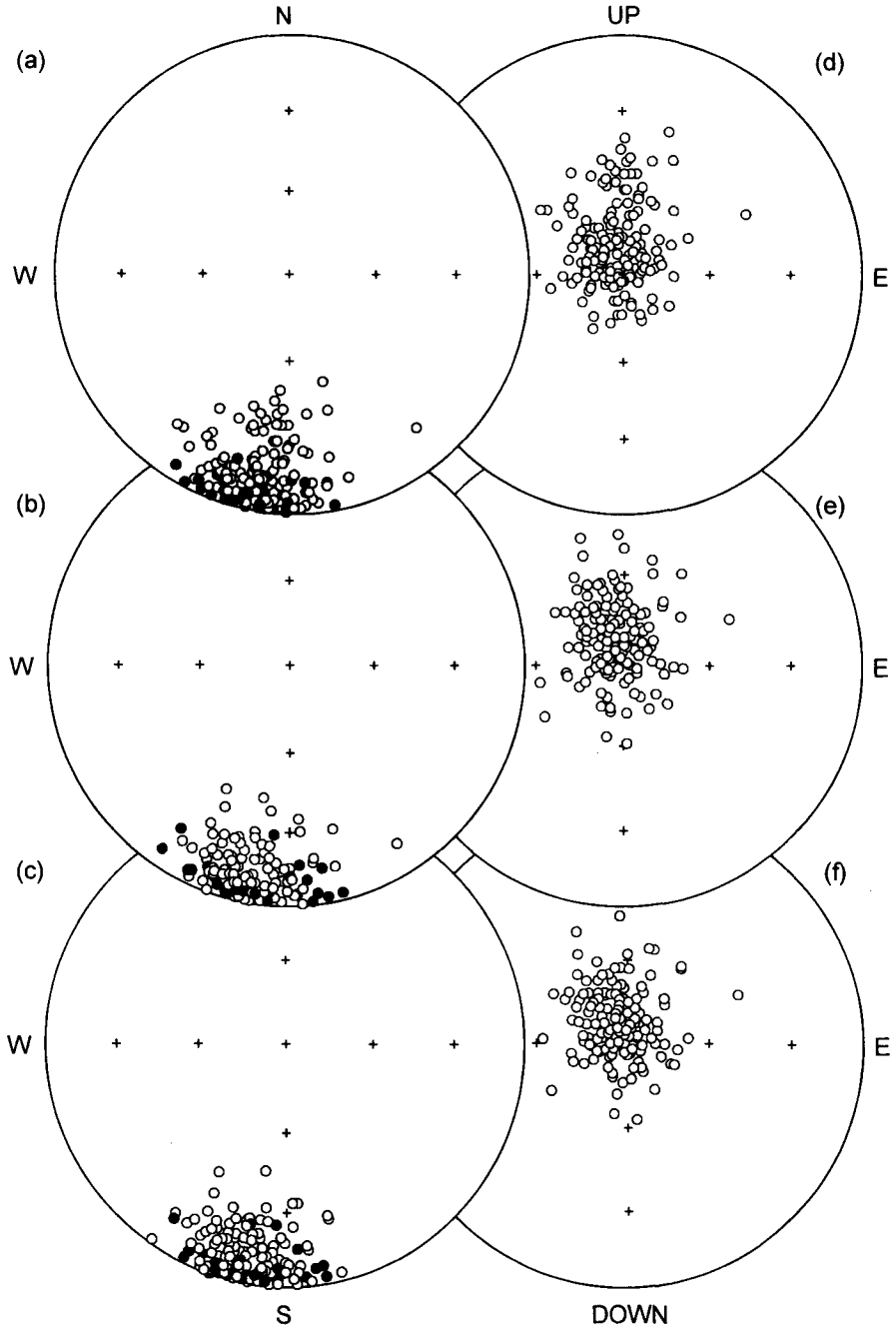


Fig. 6. Equal-area stereographic projections of high-temperature (> 650°C) component directions (structurally corrected for tectonic dip) determined for the mini-specimen fold test, showing (a) attitudes before unfolding, (b) grouping after 100% unfolding, and (c) grouping after 68% unfolding. (d–f) show the same data as (a–c), but are rotated so the mean directions are near vertical to allow better visualisation of the groupings.

the fold structure. A total of 185 mini-specimens were cored, approximately 60 from each of the steepest N-dipping and steepest S-dipping laminae and the flat-lying laminae of the block sample. The structural attitude of the bedding in each mini-specimen was carefully measured using a special goniometer. Bedding attitudes are accurate to a few degrees, and bedding planes typically vary by less than a degree throughout the volume of each mini-specimen.

Similar thermal demagnetisation steps were used as before [13], and below 300°C all specimens revealed small northerly components, presumably related to viscous remanent magnetisation in the present field direction. At higher temperatures, univectorial components were isolated which decayed towards the origin at between 650 and 685°C.

The directions of the high-temperature components were subjected to a fold test [24], again using 'definition 1' as for the regional study of the Elatina Formation. The present detailed study yielded unequivocal evidence for synfolding (very early) acquisition of remanence. The statistics are given in Table 2, and show that a minimum SCOS value (i.e., the least correlation between structure and magnetisation direction) is attained after 68% unfolding, and a maximum κ (tightest clustering) occurs at 66% unfolding. These best estimates of the palaeohorizontal are essentially identical. As a general rule, the tightest cluster does not necessarily give the best approximation to the true mean; however, with this sample, which was deposited in less than one year (see Fig. 5), no secular variation is expected so this approach is justifiable. The close agreement of the SCOS minimum and the κ maximum at 68% and 66% unfolding, respectively, confirms our earlier findings using standard-sized specimens [13] where a 67% unfolding gave the maximum κ. Fig. 6a–c shows stereographic projections of the attitudes of the high-temperature components before unfolding, after 100% unfolding and after 68% unfolding. Fig. 6d–f shows the same data as Fig. 6a–c but rotated so the mean directions are near vertical, to better display the groupings.

This soft-sediment fold test is significant at a high level of confidence. The increase in κ by a factor of 1.41 from its value before unfolding (22.9) to its maximum values (32.2, 32.3) is significant at the 99% confidence level ($F_{368,368} = 1.29$ at 99%). The increase in κ by a factor of 1.26 (28.9/22.9) from its value before unfolding to that for 100% unfolding is significant at the 95% confidence level ($F_{368,368} = 1.20$ at 95%).

The mean direction prior to unfolding has a declination of 190.0° and an inclination of −9.1° ($\alpha_{95} = 2.2°$), the mean unfolded direction has a declination of 190.0° and an inclination of −7.5° ($\alpha_{95} = 2.0°$), and the mean direction corresponding to the minimum SCOS (68% unfolded) has a declination of 190.1° and an inclination of −8.0° ($\alpha_{95} = 1.9°$) (all directions corrected for regional structure). These findings compare closely to our previous result [13], which had a maximum κ at 67% unfolding and yielded a declination of 185.2° and an inclination of −5.9° ($\alpha_{95} = 1.8°$). The slight difference between the two results probably reflects a small orientation error of the first-sampled block [13] which, although close to in situ, had been loosened from the outcrop by stream floods. The mean directions determined for the present block sample corresponding to maximum κ and minimum SCOS (Table 2) are in excellent agreement with the mean direction determined here for the three stratigraphic sections of the Elatina Formation and the mean direction determined by Embleton and Williams [12] for the rhythmites at Pichi Richi Pass (Table 1).

The present results confirm the findings of Schmidt et al. [13,14] of very early acquisition of a DRM for the rhythmites at Pichi Richi Pass. The acquisition of such a remanence is attributable to the magnetic mineralogy and special conditions of deposition—settling of haematite grains from suspension in quiet waters below storm wave base.

2.4. Drill core from the Stuart Shelf and Torrens Hinge Zone

The flat-lying, condensed Neoproterozoic sequence on the Stuart Shelf and equivalent strata in the western part of the adjacent, 15–60 km wide Torrens Hinge Zone were not deformed by the Cambro-Ordovician Delamerian Orogeny. Fifty-four samples of redbed facies, comprising mudstone, siltstone, and fine-to coarse-grained sandstone of probable continental to shallow-water marine origin, were taken from cores from six deep drill holes intersect-

ing the Elatina equivalent within a meridional area covering some 100 × 20 km of the Stuart Shelf and western Torrens Hinge Zone (Fig. 1). The core samples spanned as much as 76 m vertical thickness of the Elatina equivalent at depths ranging from 82.5 to 709 m. Bedding structures in the cores indicated very low dips and allowed the facing of samples, although the samples were unoriented in azimuth. All the cores are held at the Core Library of the South Australian Department of Mines and Energy, Adelaide, and drillhole locations are shown on the 1:600,000 geological map of the Adelaide Geosyncline and Stuart Shelf [1]. Number (see Fig. 1), name and interval sampled for each drillhole are as follows: 5, SLT-103, 82.50–104.22 m; 6, SLT-105, 475.50–529.12 m; 7, SLT-106, 531.20–553.74 m; 8, SLT-101, 313.05–388.96 m; 9, SLT-104, 361.10–439.54 m; 10, MRD-1, 704.38–709.00 m.

A total of 60 specimens were analysed for remanence following progressive AF and thermal demagnetisation. Intermediate to high inclinations probably related to late Tertiary present field directions were generally removed by heating, leaving low inclinations (usually $< 10°$) predominant at temperatures $> 650°C$ (Fig. 4i–l). Positive and negative low inclinations were present in about equal proportion. Although techniques exist for determining the mean inclination from inclination data only [28], such as from drill core, a reliable mean cannot be calculated when inclinations are low, as in the present case, because of the resulting ambiguity of polarity. The stable, low-inclination component for the flat-lying Elatina equivalent on the Stuart Shelf and western Torrens Hinge Zone may be equated with the stable, low-inclination component determined for the Elatina Formation in the Adelaide Geosyncline [12–14] and confirmed by the present study.

2.5. Summary of palaeomagnetic results

Palaeomagnetic study of outcrops of the Elatina Formation from the Central Flinders Zone of the Adelaide Geosyncline, and of drill core from flat-lying equivalent facies on the Stuart Shelf and Torrens Hinge Zone, has provided consistent results. Reddish brown sandstones, siltstones and tillite of the Elatina Formation possess a stable remanence (early CRM) of low inclination. Positive fold tests on syndepositional soft-sediment folds demonstrate that tidal rhythmites acquired a DRM very soon after deposition, through the settling of haematite grains from suspension in quiet waters. Concordant palaeomagnetic directions determined for these rhythmites and other facies of the Elatina Formation show that the formation acquired its CRM close to the time of deposition.

In view of the early timing of remanence, the existence of polarity reversals within a stratigraphic section and within some samples provides strong evidence for the identification of a dipole field axis (notwithstanding a departure from anti-parallelism, as discussed above) and the sufficient averaging of secular variation to allow the calculation of a true palaeopole for the Elatina Formation as a whole. The palaeopole derived from the sample results (structurally corrected) is located at 52.4°S, 347.1°E ($d_p = 3.7°$, $d_m = 7.4°$) and from site results (structurally corrected) at 51.5°S, 346.6°E ($d_p = 7.7°$, $d_m = 15.3°$). The overall formation mean direction determined from the sample results (structurally corrected) has a declination of 197.3° and an inclination of −5.3° ($\alpha_{95} = 7.4°$), and indicates a palaeolatitude of deposition of 2.7° ± 3.7°N. The pole position and assigned geographic polarity accord with the Neoproterozoic–Palaeozoic apparent polar wander path for Australia [11,13,18].

The present results not only highlight the equatorial palaeolatitude of deposition for these glaciogenic rocks, but also confirm earlier arguments [12–14] that the determined Elatina pole positions (albeit VGPs) were close to a true palaeomagnetic pole.

3. Discussion and conclusions

3.1. The Marinoan glacial–periglacial environment

Much evidence indicates that a frigid, strongly seasonal climate prevailed near sea level in marine to near-coastal settings during Marinoan glaciation in South Australia. The Marinoan glacial sequence in the Adelaide Geosyncline region (Fig. 1) includes thick (as much as > 1500 m) and extensive (~ 1000 km) tillite and argillite–dropstone facies containing faceted, polished and striated clasts of various lithologies including dolerite, vesicular basalt, sand-

adjacent material. A strongly seasonal climate is essential for wedge formation because only the seasonal temperature cycle, together with rapid drops of temperature, can cause the required contraction cracking. A periglacial origin of the sand-wedge polygons on the Stuart Shelf can be inferred with confidence, because they formed in quartzite breccia and quartz sand containing no clay matrix or evaporite cement; such quartzose materials do not crack and expand under dry–wet cycles [29].

Periglacial sand-wedge structures are a most reliable indicator of in-situ palaeoclimate because they form by *physical* weathering and hence their interpretation is not clouded by such uncertainties as the former nature of the atmosphere and biosphere and the effects of diagenesis. The structures imply mean annual air temperatures as low as -12 to -20°C or lower, and a seasonal temperature range as great as 40°C (mean monthly air temperatures ranging from < -35°C in mid-winter to $< +4$°C in mid-summer) [29–31,35]. The wedges on the Stuart Shelf evidently took as long as 4 kyr to form and occur in at least four generations [29–31]. Hence, a frigid, strongly seasonal climate prevailed for a substantial time in that region during the Marinoan glaciation. The presence of a strongly seasonal regime during Marinoan glaciation is supported by the powerful annual signal revealed by Fourier spectral analysis of tidal–climatic data recorded by the Elatina rhythmites [5–8,31,36].

Importantly, the Stuart Shelf was inundated by the Marinoan post-glacial marine transgression and Marinoan periglacial deposits are conformably overlain by late Neoproterozoic marine strata [1,2]. The occurrence of Marinoan glaciation near sea level in South Australia is confirmed by the presence of widespread tidal deposits in the glaciogenic Elatina Formation and its Reynella Siltstone Member, which display abundant tidal features such as herringbone cross-bedding, flaser bedding and tidal rhythmites; the Elatina–Reynella tidal rhythmites have provided detailed and self-consistent palaeotidal and palaeorotational data [5–8,36]. Indeed, detailed facies and palaeogeographic models for Marinoan glacial marine sedimentation in the Adelaide Geosyncline region [1,2] show grounded glaciers and ice sheets near sea level.

Much geological data for the Adelaide Geosyncline region therefore indicate an in-situ frigid, strongly seasonal climate near sea level in marine to near-coastal settings during Marinoan glaciation.

3.2. The Neoproterozoic climatic paradox

Palaeomagnetic data for Neoproterozoic strata in Australia and on other continents have previously suggested low palaeolatitudes for several Neoproterozoic glaciogenic sequences, although some data are equivocal or contentious (see discussions and references in [11,12,16,31]). This suggestion of glaciation in low palaeolatitudes is strongly supported by palaeomagnetic data for the Elatina Formation, which are the best constrained for any Neoproterozoic glaciogenic deposit. Indeed, the palaeomagnetic, palaeoclimatic and palaeogeophysical findings for Marinoan rocks in the Adelaide Geosyncline region form the most comprehensive body of palaeoenvironmental data currently available for a Neoproterozoic glaciogenic sequence. Together with these data, the present results confirm the Neoproterozoic climatic paradox in South Australia: *frigid strongly seasonal climate, permafrost, and grounded glaciers near sea level in equatorial palaeolatitudes* ($2.7° \pm 3.7°N$). The existence of a non-uniformitarian regional palaeowind direction (palaeonorthwesterly) with respect to low palaeolatitudes in South Australia during Marinoan glaciation, as indicated by the periglacial–aeolian Whyalla Sandstone on the Stuart Shelf (Fig. 1, inset; [31]), increases the climatic paradox. So also does the faster rotation of the Earth in the past (~ 21.9 hour day at 650–600 Ma [5–8]), which would have caused less efficient poleward transport of heat and resulting higher equatorial surface temperatures and lower polar temperatures [37]. Indeed, climate modelling of circumpolar glaciation suggests that Neoproterozoic ice sheets could not have extended closer to the equator than 25° latitude, even with a reduced solar constant and the most favourable continental configuration [38]. Hence the evidence from geology and palaeomagnetism for a frigid climate near sea level in equatorial palaeolatitudes presents palaeoclimatology with a major enigma.

The main two hypotheses for explanation of apparent low-palaeolatitude glaciation during the Neoproterozoic are Harland's [39] hypothesis of 'global

stone and dolomite [1–4]. Faceted clasts show as many as three sets of striae, providing firm evidence of glaciation. Many of the clasts can be traced to sources within the Central Flinders Zone of the Adelaide Geosyncline [3], indicating the former presence of grounded glaciers in this area. Correlative facies on the Stuart Shelf include permafrost regolith displaying primary sand-wedge polygons and other periglacial features such as frost-shattered block-field breccia of as much as 20 m in thickness, frost thrusts, truncated earth mounds, drop involutions, periglacial injections, and frost-heaved blocks [29–31]. These features of the Marinoan sequence together uniquely define glacial to periglacial environments within the Adelaide Geosyncline region. Our field observations of the Neoproterozoic succession in South Australia, and those of numerous other geologists over many decades, provide support for the conclusion of Deynoux et al. [32] that "the vast majority of documented ancient glaciogenic deposits correctly record periods of cold climate during Earth history".

Excellently preserved primary sand wedges on the Stuart Shelf (Fig. 7) are as much as 3 m deep and define polygons 10–20 m across; in their consistent wedge-form section, near-vertically laminated sand fill, and upturning of adjacent ground, the structures are identical to sand wedges forming today in the Dry Valleys of Antarctica and the arid Arctic (see [33,34]). Such polygonal sand wedges result from repeated winter contraction-cracking of the upper part of the permafrost and infilling of the cracks by windblown sand, alternating with summer thermal-expansion of the permafrost that causes upturning of



Fig. 7. Vertical section of Marinoan periglacial sand wedges at the Cattle Grid copper mine, Stuart Shelf (see Fig. 1 for locality). The large wedge (*2*) is developed in permafrost regolith of brecciated (frost-shattered) Mesoproterozoic (~ 1400 Ma) quartzite. It is 3 m deep and displays a near-vertical, diverging lamination that parallels the wedge margins. Two deformed sand wedges of an earlier generation (*1*) occur within the breccia, and a third-generation wedge (*3*) occurs within the uppermost part of the large wedge and in the overlying Marinoan periglacial–aeolian Whyalla Sandstone. Bedding in breccia and sandstone is turned upward adjacent to respective wedges. These and numerous other polygonal sand wedges formed on the Stuart Shelf near sea level during Marinoan glaciation. Identical sand wedges are forming today in the Dry Valleys of Antarctica and in the arid Arctic (see text). The structures indicate a very cold, arid, windy, and strongly seasonal climate [29,30,33–35].

(pole-to-pole) glaciation' caused by global refrigeration and Williams' [31,36,40] 'large obliquity' hypothesis of pre-Ediacaran ($\geq$ 590 Ma, see [9]) preferential low-latitude (0– $\sim$ 30°) glaciation under a strongly seasonal regime. Williams [31,36] reviewed the evidence for these and other proposed explanations of low-palaeolatitude glaciation. He concluded that the global refrigeration hypothesis is unsatisfactory for several reasons, mainly because permanent snow and ice sheets would not form at sea level in the tropics since the frozen oceans would be unable to supply the required water vapour ([41], see also [38]) and because there would be little seasonal variation with global refrigeration since the very low temperatures would inhibit precipitation and sublimation [42]. Intriguingly, however, the scenario for an obliquity of the ecliptic $> 54°$ [31,36,40] may explain the available geological, palaeomagnetic and palaeogeophysical observations for South Australia: a palaeoequator with in-situ cold climate near sea level, marked seasonality, palaeowesterly winds, and powerful semi-annual and annual signals in palaeotidal patterns [7,31,36].

The concept of a large obliquity for the Precambrian Earth is compatible with a giant impact that many argue produced the Moon (see [43,44]). Mechanisms to accomplish a proposed reduction of obliquity during the Ediacaran ($\sim$ 590–530 Ma) from $> 54°$ to a value approaching that of today [36] may include obliquity–oblateness feedback through Neoproterozoic ice-sheet loading and unloading *near the equator* (cf. Bills' [45] feedback model for *polar* ice sheets) in concert with dissipative processes at the core–mantle boundary (e.g., [36,46,47]) during a major resonance of the fluid core (see [48]). These ideas will be pursued elsewhere.

In conclusion, while the paradigm of circum-polar glaciation near sea level in the geological past remains unchallenged for the Phanerozoic, it appears contradicted by detailed geological and palaeomagnetic studies of pre-Ediacaran Neoproterozoic glaciogenic rocks in South Australia. The results of the present study confirm and highlight the Neoproterozoic climatic paradox of equatorial glaciation and frigid climate near sea level under a strongly seasonal regime. Final resolution of this paradox, which will require palaeomagnetic and geological data for other Neoproterozoic glaciogenic sequences on several continents of a quality comparable to findings for the Adelaide Geosyncline region, may illuminate the dynamics of the Precambrian Earth–Moon system and the rapid rise of metazoan life during the Ediacaran.

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